Project Scheduling with PERT and CPM, Critical Path Analysis, and Schedule Estimation

Software Engineering & Project Management • Student Learning Book

Learning outcomes

• Explain the purpose of network-based project scheduling.

• Represent activities, durations, and dependencies in a project network.

• Calculate earliest and latest times and identify the critical path using CPM.

• Use PERT's three-point estimates for uncertain activity durations.

• Explain float/slack and how delays affect the project finish date.

Running case study: A team is preparing the College Event Management System (CEMS) for release before the college festival.

1. Why Use PERT and CPM?

A task list or Gantt chart shows planned work, but complex projects also need a clear view of dependencies. Network scheduling represents which activities must happen before others, helping the team understand the shortest possible project duration under the given assumptions and which activities need close attention.

Method	Main purpose	Typical use
CPM — Critical Path Method	Uses activity durations and dependencies to calculate schedule times, float, and the critical path	Planning when durations are treated as known estimates
PERT — Program Evaluation and Review Technique	Uses optimistic, most likely, and pessimistic estimates to account for uncertainty	Planning when activity duration is uncertain

Both methods depend on a sensible activity list and accurate dependencies. They do not automatically account for every real-world constraint, such as a developer being assigned to two tasks at once, unless those constraints are modelled.

2. Basic Network Scheduling Terms

Term	Meaning
Activity	A defined piece of work with a duration
Predecessor	An activity that must be completed before another activity can begin
Successor	An activity that follows another activity
ES / Earliest Start	Earliest time an activity can start based on its predecessors
EF / Earliest Finish	Earliest Start plus duration
LS / Latest Start	Latest time an activity can start without delaying the project finish
LF / Latest Finish	Latest time an activity can finish without delaying the project finish
Float / Slack	Time an activity can be delayed without delaying the project finish, for total float
Critical path	The longest-duration path through the network; it determines the earliest possible finish under the model assumptions

3. Example Project Activities

Use these illustrative activities for a small CEMS release. Durations are in working days and are provided only for learning.

ID	Activity	Duration	Predecessor(s)
A	Confirm scope and requirements	3	None
B	Create design and UML	2	A
C	Implement event catalogue	4	B
D	Implement registration	5	B
E	Integrate modules	2	C and D
F	System testing	3	E
G	Fix defects and retest	2	F
H	Deploy and hand over	1	G

Activities C and D can begin after B. They can proceed in parallel only if suitable people and resources are available. Both must finish before E can start.

4. Forward Pass: Earliest Times

The forward pass calculates earliest start (ES) and earliest finish (EF). Start at time zero. For a simple activity-on-node schedule:

> **EF = ES + Duration**
> **ES = the largest EF among all immediate predecessors** (or 0 if there are no predecessors).

ID	Duration	ES	EF	Calculation
A	3	0	3	0 + 3
B	2	3	5	3 + 2
C	4	5	9	5 + 4
D	5	5	10	5 + 5
E	2	10	12	max(9, 10) + 2
F	3	12	15	12 + 3
G	2	15	17	15 + 2
H	1	17	18	17 + 1

The earliest finish for the project is day 18 in this example. Because E depends on both C and D, it cannot start until the later predecessor (D) has finished.

5. Backward Pass: Latest Times

The backward pass works from the project finish back toward the start. Set the final activity's LF to the project's earliest finish (18 here), then calculate:

> **LS = LF − Duration**
> **LF = the smallest LS among immediate successors**.

ID	Duration	LS	LF	Calculation
H	1	17	18	18 – 1
G	2	15	17	17 – 2
F	3	12	15	15 – 3
E	2	10	12	12 – 2

ID	Duration	LS	LF	Calculation
C	4	6	10	LS of E = 10
D	5	5	10	LS of E = 10
B	2	3	5	min(6, 5) = 5 for LF
A	3	0	3	LS of B = 3

For activity C, LF is 10 because its successor E must start no later than day 10; its LS is 10 – 4 = 6. Activity D has LS = 5. These calculations show that the schedule has flexibility on C, but not on D.

6. Float and the Critical Path

For the simple schedule shown, total float can be calculated as:

Total Float = LS – ES = LF – EF

ID	ES	EF	LS	LF	Total float
A	0	3	0	3	0
B	3	5	3	5	0
C	5	9	6	10	1
D	5	10	5	10	0
E	10	12	10	12	0
F	12	15	12	15	0
G	15	17	15	17	0
H	17	18	17	18	0

The critical path is **A → B → D → E → F → G → H**, with a total duration of **18 working days**. These activities have zero total float. Activity C has one working day of total float in this simplified network.

A delay to a zero-float activity can delay the project finish unless the team recovers time elsewhere. Critical paths can change as durations and dependencies change; they should be reviewed when the plan is updated.

7. PERT Three-Point Estimation

PERT estimates a task duration using three values:

• Optimistic (O): duration if things go particularly well.

• Most likely (M): the duration considered most plausible under normal conditions.

• Pessimistic (P): duration if significant but plausible difficulties occur.

Expected duration: TE = (O + 4M + P) / 6
Approximate variance: ((P – O) / 6)²

Example: a UML design task has O = 2 days, M = 4 days, and P = 8 days.

TE = (2 + 4×4 + 8) / 6 = 26 / 6 ≈ **4.33 days**.
Variance = ((8 – 2) / 6)² = 1 day². Standard deviation = 1 day.

These are estimates, not guarantees. The PERT formula is a simplifying model; its usefulness depends on the quality of the three estimates and the assumptions about uncertainty.

8. PERT vs. CPM

Feature	CPM	PERT
Duration input	Usually one duration estimate per activity	Three estimates: optimistic, most likely, pessimistic

Feature	CPM	PERT
Emphasis	Critical path, float, and schedule control	Expected duration and uncertainty
Best suited to	Work where duration estimates are reasonably stable	Work with meaningful uncertainty in task duration
Shared feature	Both use activity dependencies and a project network	Both use activity dependencies and a project network

9. Practical Schedule Management

• Keep activity definitions and completion criteria clear.

• Validate dependencies with the people doing the work.

• Check whether parallel activities compete for the same person or environment.

• Monitor critical-path activities and near-critical activities with little float.

• When a task slips, calculate which successors and milestones are affected.

• Discuss recovery options: reduce scope, remove a blocker, re-sequence work, or add resources if feasible.

• Update the schedule based on actual progress and document major changes.

Crashing means attempting to shorten project duration by adding resources or spending more on selected activities; it can increase cost and may not help if the activity is not on the critical path. Fast-tracking means overlapping activities that were planned sequentially; it can save time but may increase rework and coordination risk.

10. Classroom Activity: Calculate a Critical Path

Use the following small network. Assume resources are available and all dependencies are finish-to-start.

Activity	Duration (days)	Predecessors
A: Requirements	2	None
B: UI design	3	A
C: Database design	2	A
D: Build UI	4	B
E: Build API	5	C
F: Integration	2	D and E
G: Testing	3	F

1. Draw the activity network.

2. Perform a forward pass to calculate ES and EF.

3. Perform a backward pass to calculate LS and LF.

4. Calculate total float for each activity.

5. Identify the critical path and expected project duration.

6. Explain which activity delay would be most concerning and why.

Suggested check: The paths are A–B–D–F–G = 14 days and A–C–E–F–G = 14 days. Both paths are critical in this example; verify this by calculating the earliest and latest times.

11. Review Questions with Suggested Answers

1. What is the critical path?

Suggested answer: The longest-duration path through the dependency network; it determines the earliest possible finish under the model assumptions.

2. What is the forward pass used for?

Suggested answer: Calculating earliest start and finish times.

3. What is the backward pass used for?

Suggested answer: Calculating latest start and finish times without delaying the planned project finish.

4. How is total float calculated?

Suggested answer: LS – ES or LF – EF.

5. What does zero float mean?

Suggested answer: The activity has no schedule flexibility under the current network and baseline.

6. What are the three PERT estimates?

Suggested answer: Optimistic, most likely, and pessimistic.

7. Calculate PERT expected duration for O=3, M=5, P=9.

Suggested answer: (3 + 4×5 + 9) / 6 = 32/6 ≈ 5.33 time units.

8. Why can a critical path change?

Suggested answer: Updated durations, dependencies, or completed work can change which path is longest.

9. What is crashing?

Suggested answer: Trying to shorten project duration by adding resources or spending more on selected activities.

10. What is fast-tracking?

Suggested answer: Overlapping activities that were previously planned sequentially, usually with added coordination or rework risk.

12. Glossary

Term	Meaning
Network schedule	Diagram of activities and dependencies
ES / EF	Earliest start / earliest finish
LS / LF	Latest start / latest finish
Float	Allowed delay before the relevant finish date is affected
Critical path	Longest-duration path through the project network
PERT	Technique using three-point duration estimates
CPM	Critical Path Method for schedule calculations
Crashing	Reducing duration by adding resources or cost to selected activities
Fast-tracking	Overlapping activities to shorten the schedule
Baseline	Approved reference plan used to compare actual progress

13. Homework

1. Create a dependency network for a Library Management System with at least eight activities.
2. Calculate ES, EF, LS, LF, and total float for each activity.
3. Identify the critical path and project duration; state your assumptions.
4. For three activities, estimate O, M, and P, then calculate PERT expected duration and variance.
5. Explain how you would respond if a critical-path activity slipped by two days.

Next chapter: Software Cost Estimation with COCOMO, Effort, and Project Risk Management.